Solar ESS Co. Ltd.
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Tel 063 151 6694, 089 632 5829 · www.solaress.org

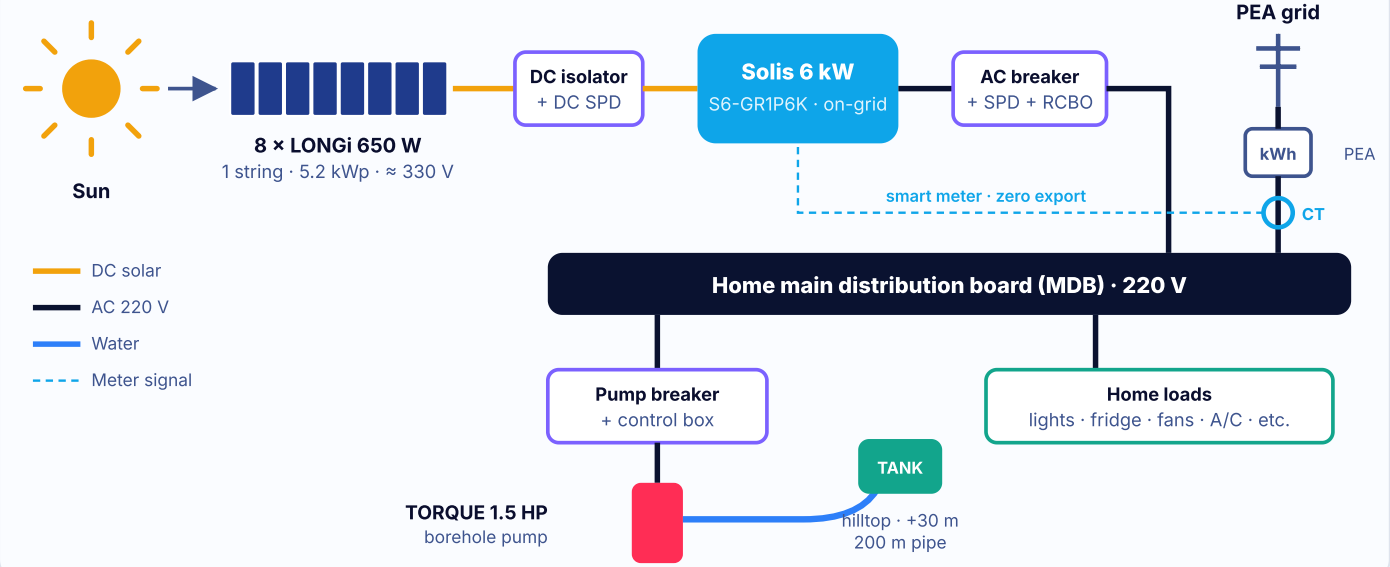
SOLAR WATER PUMPING · SYSTEM 1 · OPTION B · ON-GRID

Solar for the whole home, pump included

Instead of a dedicated solar pump inverter, 8 × LONGi 650 W panels feed a Solis 6 kW on-grid inverter connected to the home main board. The TORQUE 1.5 HP pump becomes one of the home loads, so solar power is used all day: by the pump while it fills the hilltop tank, and by the rest of the home when it is not pumping.

TN-20261019-PUMP-03-S1B | Rev01 | 20 October 2026 | Chiang Rai, Thailand | Prices in THB, confirmed at order

Single-line diagram



B On-grid system

Sun → 8 panels → Solis 6 kW on-grid inverter → home main board → pump + home loads, with zero-export metering.

5.2 kWp	6 kW	≈ 21 kWh	9 kWp
8 × 650 W, 1 string	Solis on-grid, 1-phase	Solar per sunny day	Max PV on inverter

OPTION B PRICE
144,350 THB excl. VAT
154,455 THB including 7% VAT

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OPTION A OR OPTION B?

Where the day's solar energy goes

8 × 650 W panels make about 21 kWh on a sunny day. In Option A that energy can only drive the pump; once the tank is full the rest is lost. In Option B every kWh goes to the home board, so the pump and the rest of the home share it.



SOLAR PER SUNNY DAY	PUMP USE	VALUE · OPTION A	VALUE · OPTION B
≈ 21 kWh	**≈ 9 kWh**	**≈ 15k THB/yr**	**up to 34k THB/yr**
5.2 kWp × 4.0 h	1.5 kW × 6 h	pump energy only	if the home uses it all

Side by side

	OPTION A · SOLAR PUMP INVERTER	OPTION B · ON-GRID SOLIS 6 KW
How the pump is powered	Solar first, grid backup through the Novem pump inverter	From the home board: grid and solar together
Solar energy when the tank is full	Not used: the panels only drive the pump	Used by lights, fridge, fans and A/C
Pumping at night / cloudy days	Yes, on grid backup	Yes, on grid
During a power cut	Can still pump in daylight on solar	Stops: on-grid inverters switch off for safety (anti-islanding)
PEA approval	Not needed (no connection for solar export)	Needed: Solar ESS files the PEA application with an engineer-certified SLD
Monitoring	Inverter display	Solis app: daily kWh, alarms, history
Room to grow	Pump only	Inverter accepts up to 9 kWp: about 5 more panels later
Price excl. VAT	**120,850 THB**	**144,350 THB**

Option B costs about 23,500 THB more, but can put up to 12 kWh more solar to work each day (≈ 19k THB a year), so the difference can pay for itself in about 15 months if the home uses that power in daytime. Option A is the better choice if the pump must keep running during power cuts.

Assumptions & limits

≈ Energy figures assume 4.0 kWh per kWp per day (Chiang Rai average), the pump drawing ≈ 1.5 kW for about 6 hours, and grid power at ≈ 4.5 THB/kWh. Savings depend on how much power the home uses in daytime.

! In a power cut the on-grid inverter switches off, so the pump stops until the grid returns. If pumping during outages matters, keep Option A, or add a hybrid inverter with battery later.

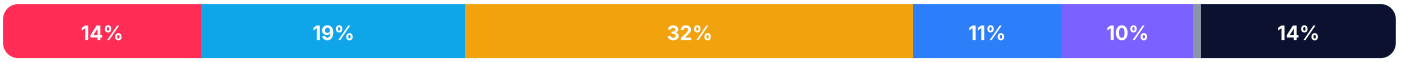
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OPTION B · BILL OF MATERIALS

On-grid system · price

TOTAL EXCL. VAT	VAT 7%	TOTAL INCL. VAT	PEA APPLICATION
144,350 THB	**10,105** THB	**154,455** THB	**Included** engineer-certified SLD

Where the money goes



Pump & control	20,500	On-grid inverter & meter	27,500	Solar array & frame	46,400	Piping & valves	15,300
Electrical & protection	13,850	Safety accessories	800	Installation	20,000		

Itemised list

#	ITEM	QTY	UNIT (THB)	AMOUNT (THB)
1	TORQUE TQ-SP-2DLBM20-4/S borehole pump, 1.1 kW (1.5 HP), 1-phase 220 V, 1½" outlet, 20 impellers, for 4" well	1 set	17,000	**17,000**
2	Pump control box: start capacitor, overload and dry-run protection, float-switch input	1 set	3,500	**3,500**
3	Solis S6-GR1P6K on-grid inverter 6 kW, 1-phase 220 V, 2 MPPT, with WiFi monitoring (budget price)	1	24,000	**24,000**
4	Solis smart meter + CT for zero export (export power manager)	1 set	3,500	**3,500**
5	LONGi 650 W solar panel	8	3,800	**30,400**
6	Galvanised ground-mount frame, 8 panels	1 set	16,000	**16,000**
7	HDPE pipe OD 50 mm PN10 (well drop + 200 m uphill line)	240 m	45	**10,800**
8	HDPE fittings, 1½"→OD50 adaptor, check valve, gate valve	1 lot	4,500	**4,500**
9	Submersible pump cable 4 × 2.5 mm²	50 m	55	**2,750**
10	PV cable 4 mm² red/black + MC4 connectors	1 lot	2,000	**2,000**
11	DC isolator + DC SPD + steel enclosure	1 set	3,500	**3,500**
12	AC side: inverter breaker + AC SPD + RCBO + enclosure	1 set	3,000	**3,000**
13	Pump circuit breaker in home main board	1	800	**800**
14	Earth rod and earth cable	1 set	1,200	**1,200**
15	Tank float switch	1	600	**600**
16	Stainless safety rope + pump clamps + well cap	1 set	800	**800**
17	Installation, pipe laying and commissioning	1 lot	20,000	**20,000**
	Total (excluding VAT)			**144,350**
	VAT 7%			10,105
	Total including VAT			**154,455**

Notes

Prices are budget figures excluding VAT unless stated. The Solis inverter price is a budget figure to be confirmed with our supplier; other items are market estimates. Final prices are confirmed at order.

An on-grid system must be approved by PEA before it is switched on. Solar ESS processes the application and supplies an SLD certified by a qualified electrical engineer (included). Selling surplus power back to PEA is an optional extra (2,140 THB application, if PEA allows); otherwise the smart meter keeps export at zero.

One string of 8 × 650 W gives ≈ 330 V at maximum power (the inverter's rated 330 V) and ≈ 395 V open-circuit, well under its 600 V limit. The panel current will be checked against the inverter's 14 A per MPPT; any excess is clipped, not a safety issue.

All electrical work uses copper cable, proper DC and AC protection and earthing, installed by a certified electrician.